NYISO 2023

Annual cost US$ 8.030 billion

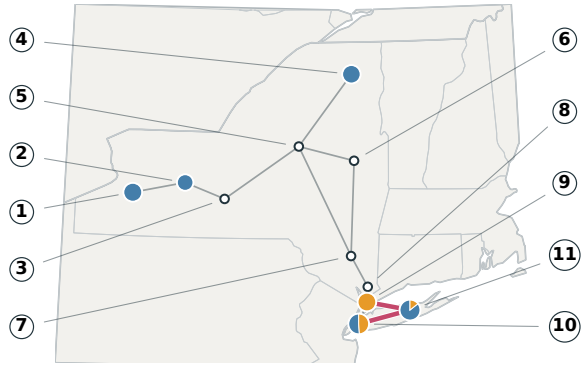
Research budget basis RGGI

GPT-6.1 Sol single-agent

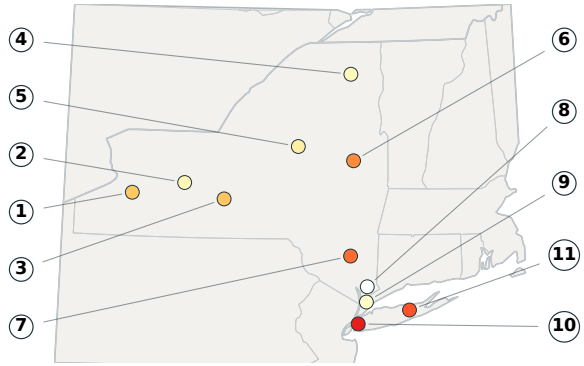
CO_2 23.479 / 24.762 Mt budget

94.82% of the system budget used

a Optimized additions



b Zonal direct CO_2



Wind Solar Nuclear Battery Existing link Added transfer

CO_2 per zone (Mt yr^{-1})

0.01 0.1 1 10 100 200

#	Zone	Existing GW	Wind GW	Solar GW	Nuclear GW	Battery GW	Energy GWh	Direct CO_2 Mt yr^{-1}
1	ZONA	3.94	3.02	—	—	—	—	0.198
2	ZONB	1.06	0.22	—	—	—	—	0.018
3	ZONC	7.30	—	—	—	—	—	0.221
4	ZOND	2.08	2.48	—	—	—	—	0.017
5	ZONE	1.92	—	—	—	—	—	0.030
6	ZONF	3.21	—	—	—	—	—	1.469
7	ZONG	6.86	—	—	—	—	—	2.655
8	ZONH	0.01	—	—	—	—	—	0.000
9	ZONI	0.06	—	2.75	—	—	—	0.005
10	ZONJ	10.76	2.82	2.70	—	—	—	14.515
11	ZONK	5.82	4.93	0.82	—	—	—	4.352
	Total	**43.00**	**13.47**	**6.27**	**—**	**—**	**—**	**23.479**

Added transfer 2.00 GW

Lowest recorded cost among four accepted trajectories.

Existing is inherited generation; additions include battery power and energy.

Emission points represent 8,760-hour dispatch totals at model-zone locations.

One common logarithmic colour scale; hollow points indicate zero emissions. Research budgets are system-wide.

NYISO 2030

Annual cost US$ 6.562 billion

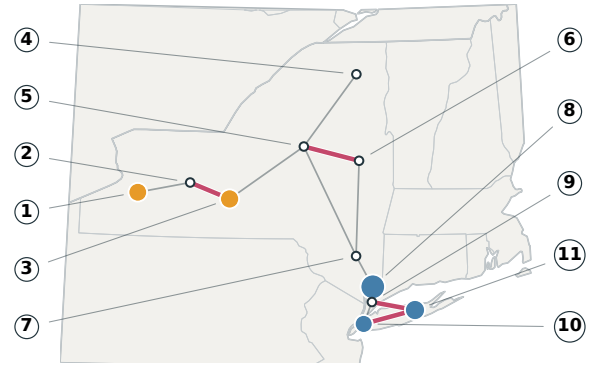
Research budget basis RGGI

GPT-6.1 Sol single-agent

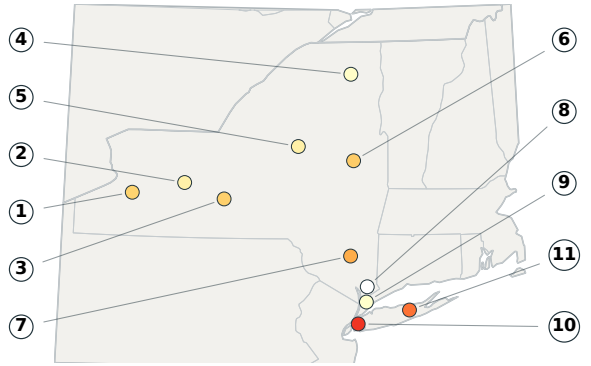
CO_2 12.425 / 12.425 Mt budget

100.00% of the system budget used

a Optimized additions



b Zonal direct CO_2



Wind Solar Nuclear Battery Existing link Added transfer

CO_2 per zone (Mt yr^{-1})

0.01 0.1 1 10 100 200

#	Zone	Existing GW	Wind GW	Solar GW	Nuclear GW	Battery GW	Energy GWh	Direct CO_2 Mt yr^{-1}
1	ZONA	3.94	—	2.80	—	—	—	0.142
2	ZONB	1.06	—	—	—	—	—	0.027
3	ZONC	7.30	—	3.36	—	—	—	0.157
4	ZOND	2.08	—	—	—	—	—	0.012
5	ZONE	1.92	—	—	—	—	—	0.029
6	ZONF	3.21	—	—	—	—	—	0.179
7	ZONG	6.86	—	—	—	—	—	0.493
8	ZONH	0.01	11.43	—	—	—	—	0.000
9	ZONI	0.06	—	—	—	—	—	0.002
10	ZONJ	10.76	1.88	—	—	—	—	8.982
11	ZONK	5.82	4.56	—	—	—	—	2.401
	Total	**43.00**	**17.86**	**6.16**	**—**	**—**	**—**	**12.425**

Added transfer 4.00 GW

Lowest recorded cost among four accepted trajectories.

Existing is inherited generation; additions include battery power and energy.

Emission points represent 8,760-hour dispatch totals at model-zone locations.

One common logarithmic colour scale; hollow points indicate zero emissions. Research budgets are system-wide.

NYISO 2035

Annual cost US$ 8.439 billion

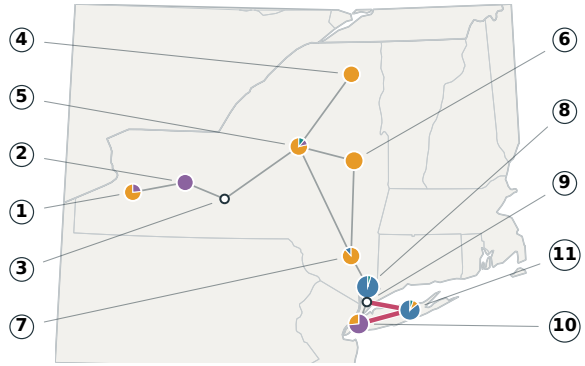
Research budget basis RGGI

GPT-6.1 Sol single-agent

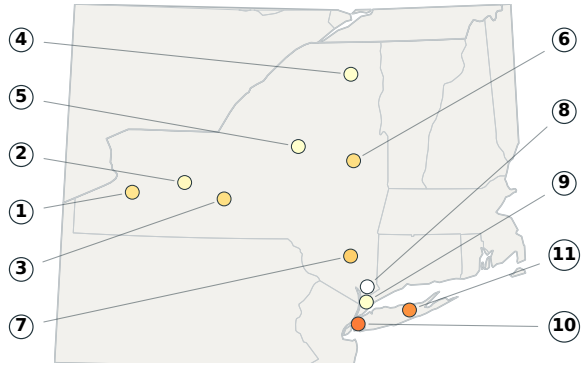
CO_2 3.550 / 3.550 Mt budget

100.00% of the system budget used

a Optimized additions



b Zonal direct CO_2



Wind Solar Nuclear Battery Existing link Added transfer

CO_2 per zone (Mt yr^{-1})

0.01 0.1 1 10 100 200

#	Zone	Existing GW	Wind GW	Solar GW	Nuclear GW	Battery GW	Energy GWh	Direct CO_2 Mt yr^{-1}
1	ZONA	3.94	—	0.86	0.27	—	—	0.058
2	ZONB	1.06	—	—	0.70	—	—	0.015
3	ZONC	7.30	—	—	—	—	—	0.079
4	ZOND	2.08	—	1.10	—	—	—	0.005
5	ZONE	1.92	—	1.86	0.25	0.26	1.03	0.008
6	ZONF	3.21	—	2.90	—	—	—	0.087
7	ZONG	6.86	0.27	2.03	—	—	—	0.167
8	ZONH	0.01	8.60	—	—	0.42	1.67	0.000
9	ZONI	0.06	—	—	—	—	—	0.000
10	ZONJ	10.76	—	1.44	4.02	<0.01	<0.01	1.939
11	ZONK	5.82	5.08	0.53	—	0.31	1.24	1.191
	Total	**43.00**	**13.95**	**10.73**	**5.24**	**0.99**	**3.95**	**3.550**

Added transfer 2.00 GW

Representative of 4 trajectories tied at the lowest recorded cost.

Existing is inherited generation; additions include battery power and energy.

Emission points represent 8,760-hour dispatch totals at model-zone locations.

One common logarithmic colour scale; hollow points indicate zero emissions. Research budgets are system-wide.

MISO 2030

Annual cost US$ 47.899 billion

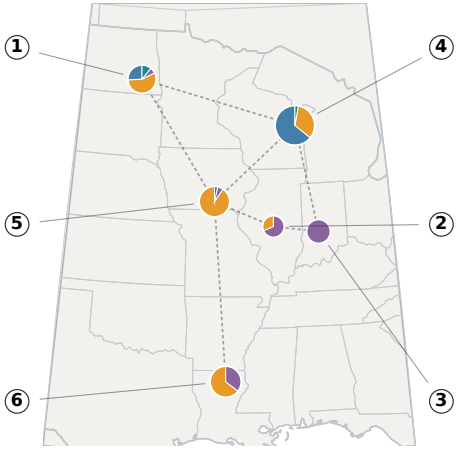
Research budget basis MISO Future 2

GPT-6.1 Sol single-agent

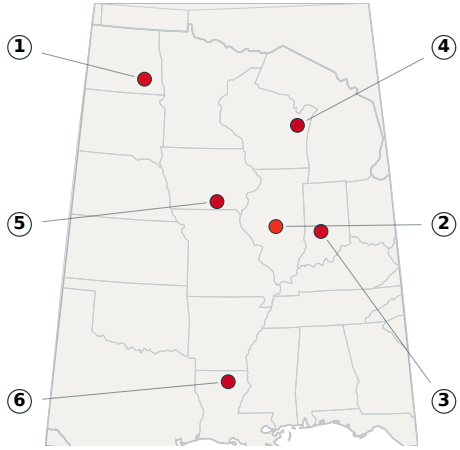
CO_2 200.033 / 200.033 Mt budget

100.00% of the system budget used

a Optimized additions



b Zonal direct CO_2



Wind Solar Nuclear Battery Existing link Added transfer

CO_2 per zone (Mt yr^{-1})

0.01 0.1 1 10 100 200

#	Zone	Existing GW	Wind GW	Solar GW	Nuclear GW	Battery GW	Energy GWh	Direct CO_2 Mt yr^{-1}
1	LRZ 1	27.88	4.82	10.47	1.25	2.00	4.00	26.947
2	LRZ 4	15.37	—	2.28	4.93	—	—	9.881
3	LRZ 6	21.54	—	—	10.58	—	—	32.132
4	LRZ 2+7	45.09	28.61	14.69	—	1.25	2.50	46.526
5	LRZ 3+5	31.78	—	20.68	1.41	0.77	1.53	38.914
6	LRZ 8+9+10	54.80	—	15.61	8.56	—	—	45.633
	Total	**196.46**	**33.43**	**63.74**	**26.73**	**4.01**	**8.03**	**200.033**

Added transfer 0.00 GW

Representative of 2 trajectories tied at the lowest recorded cost.

Existing is inherited generation; additions include battery power and energy.

Emission points represent 8,760-hour dispatch totals at model-zone locations.

One common logarithmic colour scale; hollow points indicate zero emissions. Research budgets are system-wide.

MISO 2035

Annual cost US$ 55.960 billion

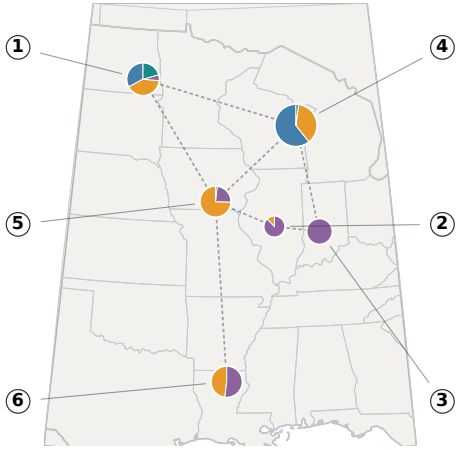
Research budget basis MISO Future 2

GPT-6.1 Sol single-agent

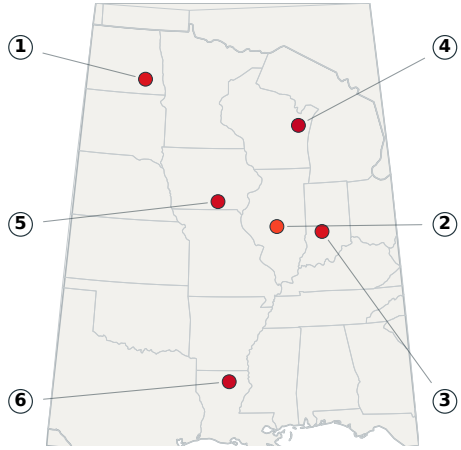
CO_2 164.642 / 164.642 Mt budget

100.00% of the system budget used

a Optimized additions



b Zonal direct CO_2



Wind Solar Nuclear Battery Existing link Added transfer

CO_2 per zone (Mt yr^{-1})



#	Zone	Existing GW	Wind GW	Solar GW	Nuclear GW	Battery GW	Energy GWh	Direct CO_2 Mt yr^{-1}
1	LRZ 1	27.88	9.26	11.46	1.65	5.80	13.34	21.722
2	LRZ 4	15.37	—	0.83	6.01	—	—	6.161
3	LRZ 6	21.54	—	—	13.92	—	—	23.771
4	LRZ 2+7	45.09	31.80	19.53	—	1.05	2.41	45.288
5	LRZ 3+5	31.78	—	18.22	5.81	0.43	1.00	31.194
6	LRZ 8+9+10	54.80	—	12.03	12.99	—	—	36.506
	Total	**196.46**	**41.06**	**62.07**	**40.38**	**7.28**	**16.75**	**164.642**

Added transfer 0.00 GW

Lowest recorded cost among four accepted trajectories.

Existing is inherited generation; additions include battery power and energy.

Emission points represent 8,760-hour dispatch totals at model-zone locations.

One common logarithmic colour scale; hollow points indicate zero emissions. Research budgets are system-wide.

Mexico 2030

Annual cost US$ 22.258 billion

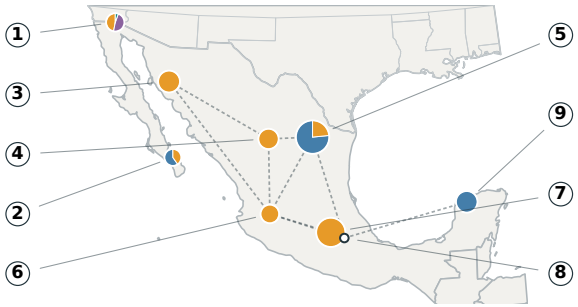
Research budget basis PROSENER

GPT-6.1 Sol single-agent

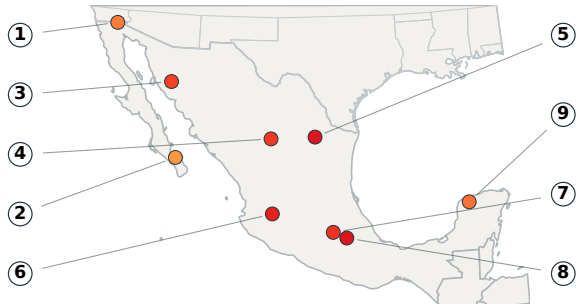
CO_2 85.664 / 119.000 Mt budget

71.99% of the system budget used

a Optimized additions



b Zonal direct CO_2



Wind Solar Nuclear Battery Existing link Added transfer

CO_2 per zone (Mt yr^{-1})

0.01 0.1 1 10 100 200

#	Zone	Existing GW	Wind GW	Solar GW	Nuclear GW	Battery GW	Energy GWh	Direct CO_2 Mt yr^{-1}
1	BCA	3.26	—	1.13	1.14	0.13	0.46	2.008
2	BCS	1.21	0.45	0.30	—	—	—	0.972
3	NOR	9.12	—	7.18	—	—	—	6.797
4	NTE	6.66	—	4.95	—	—	—	7.968
5	NES	19.36	22.02	6.73	—	—	—	21.039
6	OCC	17.21	—	2.74	—	—	—	14.000
7	CEN	6.43	—	19.83	—	—	—	9.243
8	ORI	20.05	—	—	—	—	—	21.291
9	PEN	2.80	6.74	—	—	—	—	2.347
	Total	**86.12**	**29.21**	**42.86**	**1.14**	**0.13**	**0.46**	**85.664**

Added transfer 0.00 GW

Lowest recorded cost among four accepted trajectories.

Existing is inherited generation; additions include battery power and energy.

Emission points represent 8,760-hour dispatch totals at model-zone locations.

One common logarithmic colour scale; hollow points indicate zero emissions. Research budgets are system-wide.

Brazil 2030

Annual cost US$ 24.729 billion

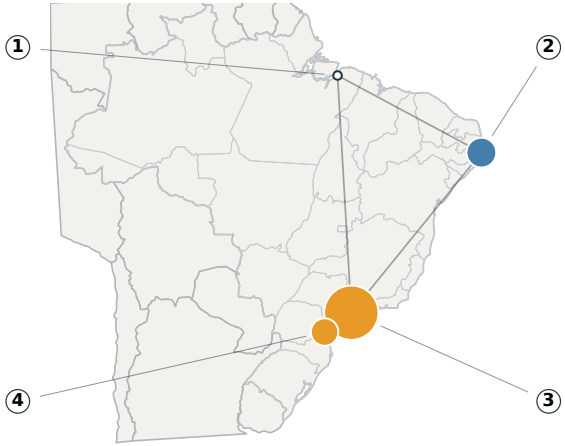
Research budget basis PDE 2035 SIN

GPT-6.1 Sol single-agent

CO_2 27.000 / 27.000 Mt budget

100.00% of the system budget used

a Optimized additions



b Zonal direct CO_2



Wind Solar Nuclear Battery Existing link Added transfer

CO_2 per zone (Mt yr^{-1})



#	Zone	Existing GW	Wind GW	Solar GW	Nuclear GW	Battery GW	Energy GWh	Direct CO_2 Mt yr^{-1}
1	N	27.24	—	—	—	—	—	3.332
2	NE	49.64	20.92	—	—	—	—	5.151
3	SE	82.46	—	90.42	—	—	—	6.939
4	S	21.11	—	16.39	—	—	—	11.577
	Total	**180.45**	**20.92**	**106.81**	**—**	**—**	**—**	**27.000**

Added transfer 0.00 GW

Representative of 4 trajectories tied at the lowest recorded cost.

Existing is inherited generation; additions include battery power and energy.

Emission points represent 8,760-hour dispatch totals at model-zone locations.

One common logarithmic colour scale; hollow points indicate zero emissions. Research budgets are system-wide.

Brazil 2035

Annual cost US$ 33.737 billion

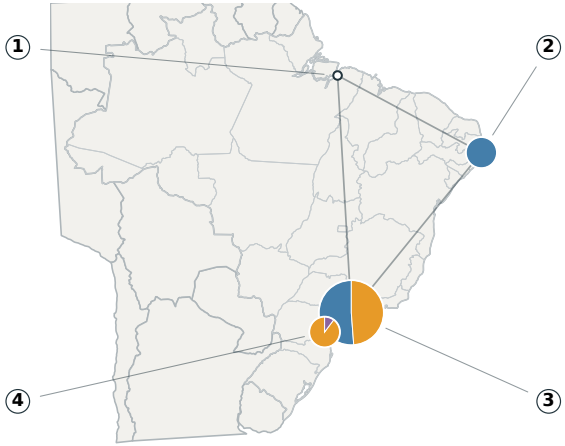
Research budget basis PDE 2035 SIN

GPT-6.1 Sol single-agent

CO_2 29.000 / 29.000 Mt budget

100.00% of the system budget used

a Optimized additions



b Zonal direct CO_2



Wind Solar Nuclear Battery Existing link Added transfer

CO_2 per zone (Mt yr^{-1})



#	Zone	Existing GW	Wind GW	Solar GW	Nuclear GW	Battery GW	Energy GWh	Direct CO_2 Mt yr^{-1}
1	N	27.24	—	—	—	—	—	3.394
2	NE	49.64	23.77	—	—	—	—	6.054
3	SE	82.46	68.69	65.51	—	—	—	7.255
4	S	21.11	—	20.52	2.33	—	—	12.298
	Total	**180.45**	**92.46**	**86.03**	**2.33**	**—**	**—**	**29.000**

Added transfer 0.00 GW

Representative of 4 trajectories tied at the lowest recorded cost.

Existing is inherited generation; additions include battery power and energy.

Emission points represent 8,760-hour dispatch totals at model-zone locations.

One common logarithmic colour scale; hollow points indicate zero emissions. Research budgets are system-wide.

South Africa 2030

Annual cost US$ 12.019 billion

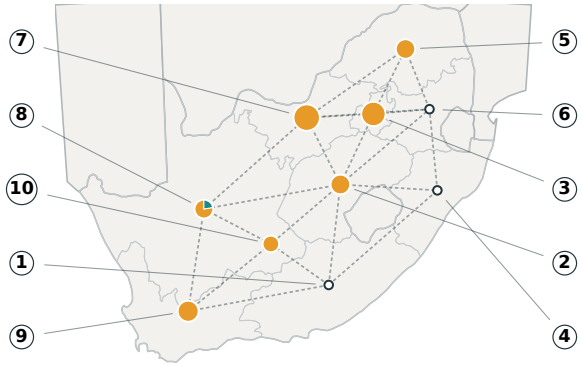
Research budget basis IRP 2025

GPT-6.1 Sol single-agent

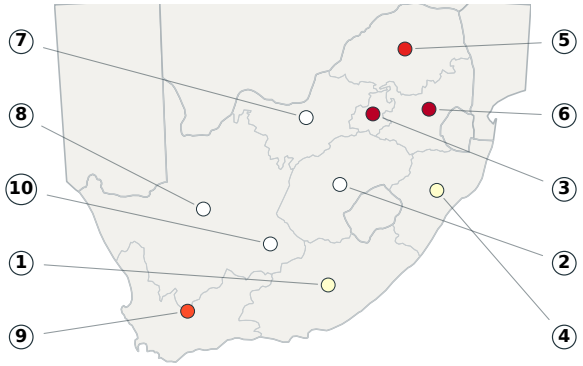
CO_2 143.655 / 168.000 Mt budget

85.51% of the system budget used

a Optimized additions



b Zonal direct CO_2



Wind Solar Nuclear Battery Existing link Added transfer

CO_2 per zone (Mt yr^{-1})



#	Zone	Existing GW	Wind GW	Solar GW	Nuclear GW	Battery GW	Energy GWh	Direct CO_2 Mt yr^{-1}
1	EC	1.82	—	—	—	—	—	0.001
2	FS	0.33	—	4.04	—	—	—	0.000
3	GP	16.95	—	11.00	—	—	—	60.501
4	KZN	1.01	—	—	—	—	—	0.011
5	LP	3.84	—	3.83	—	—	—	14.200
6	MP	14.89	—	—	—	—	—	63.936
7	NW	—	—	14.50	—	—	—	0.000
8	NC	1.81	—	2.31	—	0.66	0.66	0.000
9	WC	8.61	—	5.85	—	—	—	5.006
10	HYDRA	1.92	—	0.33	—	—	—	0.000
	Total	**51.17**	**—**	**41.85**	**—**	**0.66**	**0.66**	**143.655**

Added transfer 0.00 GW

Lowest recorded cost among four accepted trajectories.

Existing is inherited generation; additions include battery power and energy.

Emission points represent 8,760-hour dispatch totals at model-zone locations.

One common logarithmic colour scale; hollow points indicate zero emissions. Research budgets are system-wide.

South Africa 2035

Annual cost US$ 14.365 billion

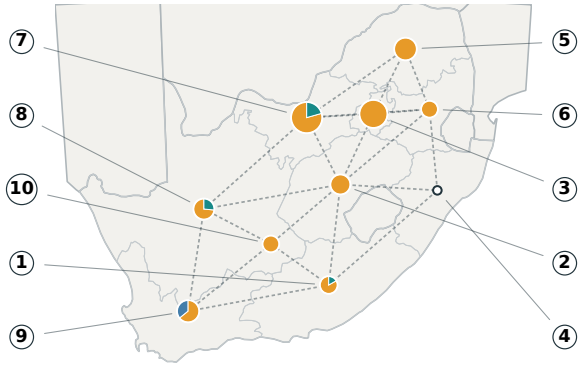
Research budget basis IRP 2025

GPT-6.1 Sol single-agent

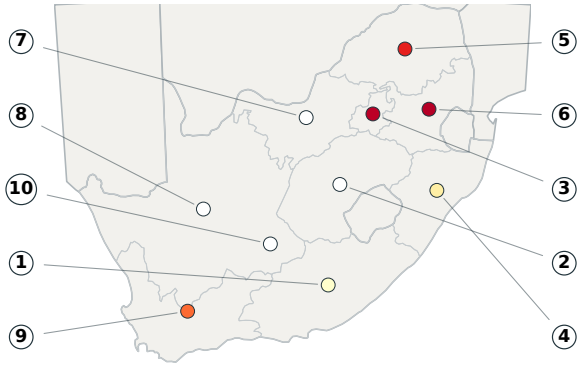
CO_2 142.000 / 142.000 Mt budget

100.00% of the system budget used

a Optimized additions



b Zonal direct CO_2



Wind Solar Nuclear Battery Existing link Added transfer

CO_2 per zone (Mt yr^{-1})



#	Zone	Existing GW	Wind GW	Solar GW	Nuclear GW	Battery GW	Energy GWh	Direct CO_2 Mt yr^{-1}
1	EC	1.82	—	1.33	—	0.27	0.80	0.001
2	FS	0.33	—	4.50	—	—	—	0.000
3	GP	16.95	—	17.57	—	—	—	59.328
4	KZN	1.01	—	—	—	—	—	0.030
5	LP	3.84	—	8.33	—	—	—	14.538
6	MP	14.89	—	0.60	—	—	—	65.352
7	NW	—	—	19.00	—	4.93	14.79	0.000
8	NC	1.81	—	4.30	—	1.55	4.64	0.000
9	WC	8.61	2.76	4.94	—	—	—	2.751
10	HYDRA	1.92	—	0.48	—	—	—	0.000
	Total	**51.17**	**2.76**	**61.05**	**—**	**6.74**	**20.23**	**142.000**

Added transfer 0.00 GW

Lowest recorded cost among four accepted trajectories.

Existing is inherited generation; additions include battery power and energy.

Emission points represent 8,760-hour dispatch totals at model-zone locations.

One common logarithmic colour scale; hollow points indicate zero emissions. Research budgets are system-wide.